Microsoft Certified Azure Administrator: AZ-104

©Simplilearn. All rights reserved.

simplilearn

TECHNOLOGY

Implement and Manage Storage

©Simplilearn. All rights reserved.

simplilearn

A Day in the Life of an Azure Architect

An Azure Administrator has diverse requirements for their cloud-hosted data. The company has documents, spreadsheets, and videos. This information needs to be securely shared across the organization and geographical areas.

An appropriate storage account for the data is to be established. It is required to configure secure access and a storage replication strategy.

The company has a large repository of documents used across the company. The offices are in different geographical regions but need the most current versions of the documents. The Azure File Shares are configured to provide a central location for the documents.

©Simplilearn. All rights reserved.

Learning Objectives

By the end of this lesson, you will be able to:

- Create and configure storage accounts
- Determine replication strategies
- Configure Azure Files
- Configure storage with tools

©Simplilearn. All rights reserved.

TECHNOLOGY

Creating and Configuring Storage Accounts

©Simplilearn. All rights reserved.

simplilearn

Features of Azure Storage

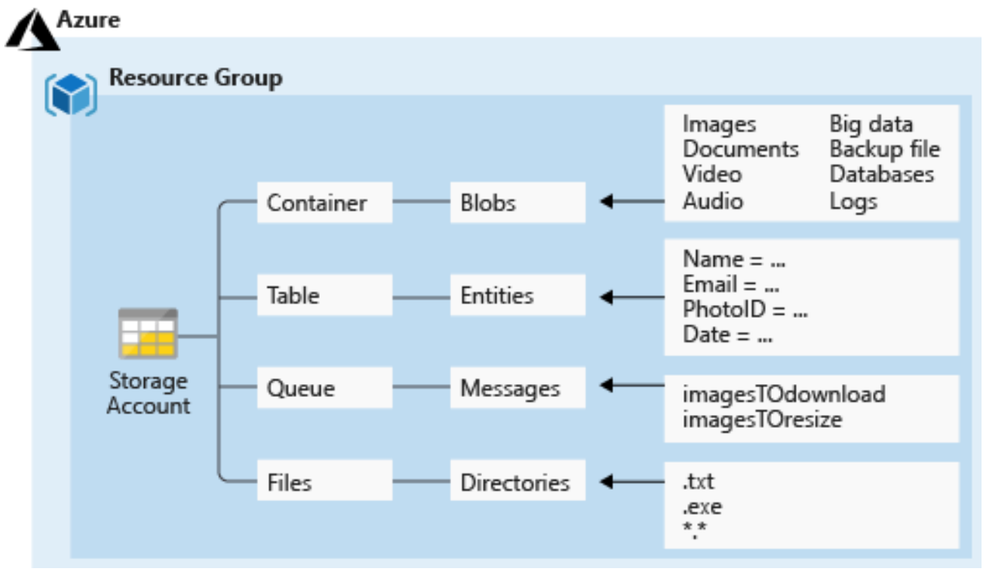
Features of Azure Storage:

- It is a service that can be used to store files, messages, tables, and other types of information.
- It is durable, secure, scalable, managed, and accessible.
- It provides storage for virtual machines, unstructured data, and structured data.
- It has two tiers: Premium and Standard.

©Simplilearn. All rights reserved.

Explore Azure Storage Services

A Microsoft object storage solution for the cloud, optimized for storing massive amounts of unstructured data through services mentioned below:



- Azure Containers: A massively scalable object store for text and binary data
- Azure Tables: Ideal for storing structured, non-relational data
- Azure Queues: A messaging store for reliable messaging between application components
- Azure Files: Managed file shares for cloud or on-premises deployments

Source: https://docs.microsoft.com/en-us/azure/?product=popular

©Simplilearn. All rights reserved.

Storage Account Types

Azure Storage offers several types of storage accounts:

Storage Account	Recommended usage
Standard general-purpose v2	Most scenarios including Blob, File, Queue, Table, and Data Lake Storage
Premium block blobs	Block blob scenarios with high transaction rates or scenarios that use smaller objects or require consistently low-storage latency
Premium file shares	Enterprise or high-performance file share applications
Premium page blobs	Premium high-performance page blob scenarios

©Simplilearn. All rights reserved.

Determine Replication Strategies

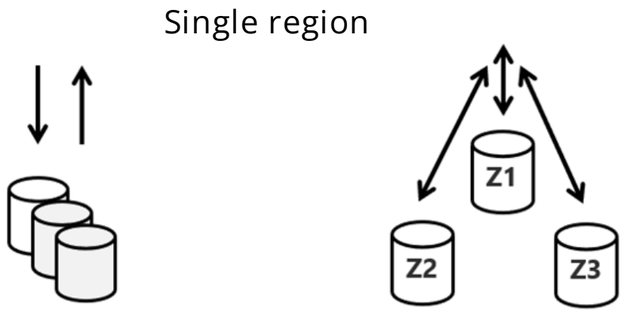
Data replication in Azure storage accounts ensures durability and high availability. Ensure data replication in the following ways:

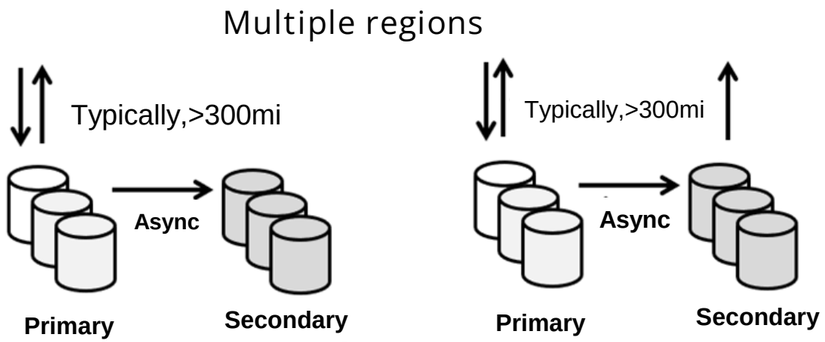
LRS

- Three replicas, one region
- Protects against disk, node, and rack failures
- Write is acknowledged with committed replicas
- Superior to dual-parity RAID

ZRS

- Three replicas, three zones, and one region
- Protects against disk, node, rack, and zone failures
- Synchronous writes to all three zones





Source: https://docs.microsoft.com/en-us/azure/?product=popular

©Simplilearn. All rights reserved.

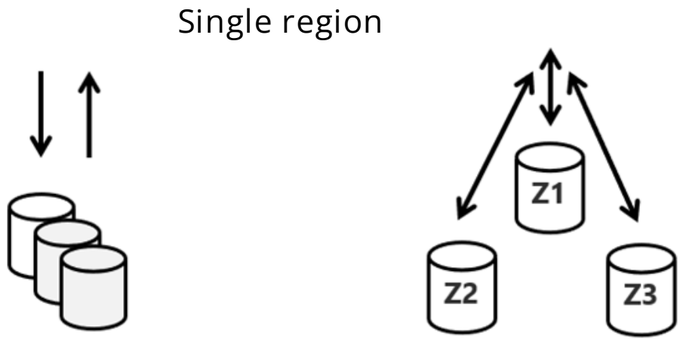
Determine Replication Strategies

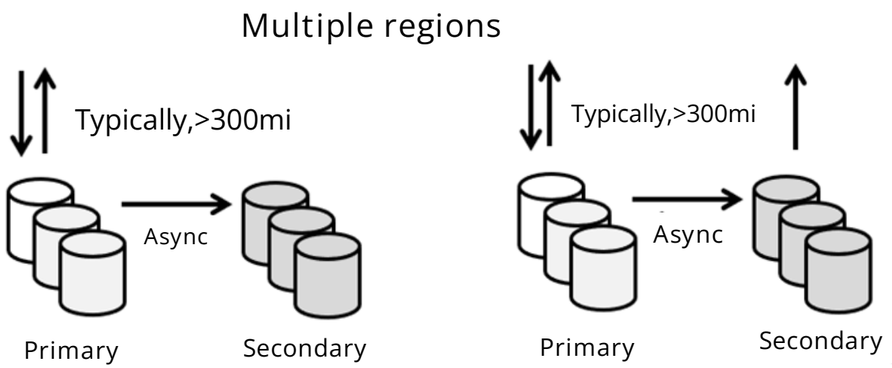
GRS

- Six replicas, two regions (Three per region)
- Protects against major regional disasters
- Asynchronous copy to secondary

RA-GRS

- GRS + read access to secondary
- Separate secondary endpoint
- Recovery point objective (RPO) delay to secondary can be queried





Source: https://docs.microsoft.com/en-us/azure/?product=popular

©Simplilearn. All rights reserved.

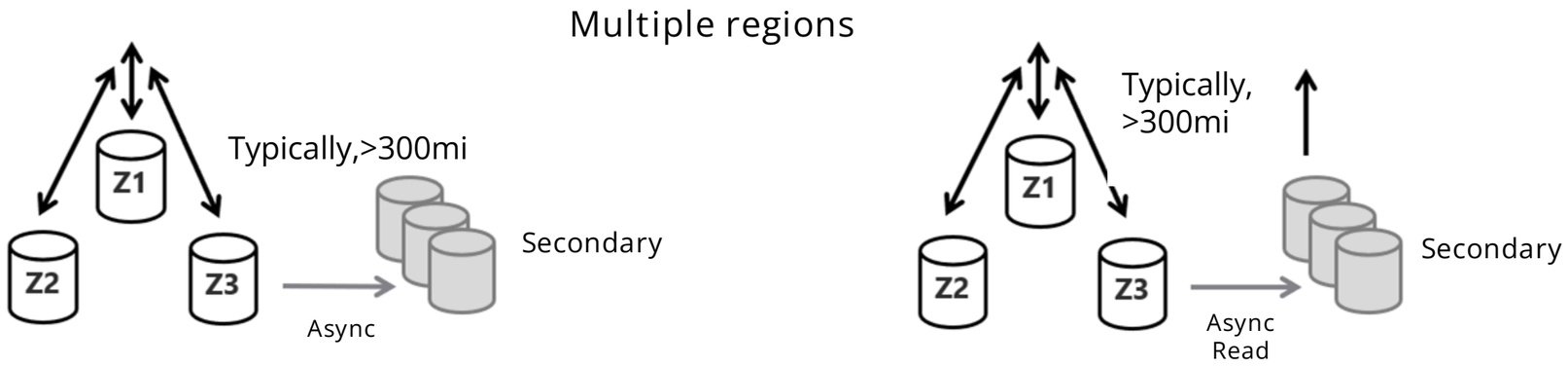
Determine Replication Strategies

GZRS

- Six replicas, 3+1 zones, and two regions
- Protects against disk, node, rack, zone, and region failures
- Synchronous writes to all three zones and asynchronous copy to secondary

RA-GZRS

- GRS + read access to secondary
- Separate secondary endpoint
- Recovery point objective (RPO) delay to secondary can be queried



Source: https://docs.microsoft.com/en-us/azure/?product=popular

©Simplilearn. All rights reserved.

Creating a Storage Account Using Azure Portal

Duration: 10 Min.

Problem Statement:

Create a storage account using the Azure portal to use different storage capabilities by Azure.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

Steps to create a storage account:

1. Sign in to Azure Portal
2. Locate storage account blade
3. Click on create storage account

©Simplilearn. All rights reserved.

Access Storage

Every object has a unique URL address based on account name and storage type.

- Container service: https://*mystorageaccount*.blob.core.windows.net
- Table service: https://*mystorageaccount*.table.core.windows.net
- Queue service: https://*mystorageaccount*.queue.core.windows.net
- File service: https://*mystorageaccount*.file.core.windows.net

A custom domain name can be configured.

An example:

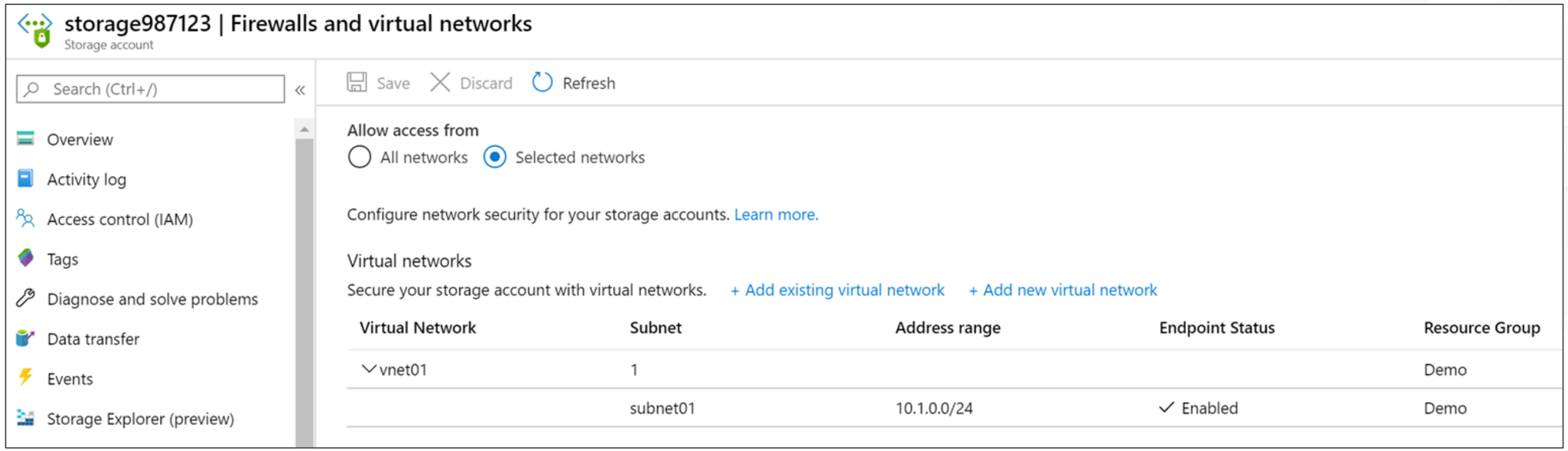
CNAME record	Target
blobs.contoso.com	contosoblobs.blob.core.windows.net

©Simplilearn. All rights reserved.

Secure Storage Endpoints

These are the necessary steps used to restrict network access to Azure services:

- Firewalls and virtual networks restrict access to the storage account from specific subnets on virtual networks or public IPs.
- Subnets and virtual networks must exist in the same Azure region or region pair as the storage account.

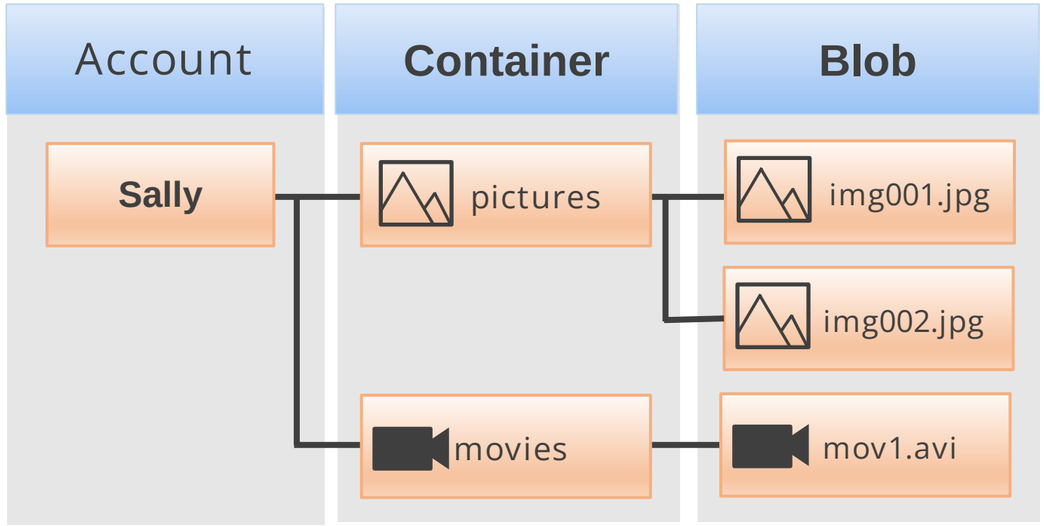


Source: https://docs.microsoft.com/en-us/azure/?product=popular

©Simplilearn. All rights reserved.

Implement Blob Storage

Blob Storage stores unstructured data in the cloud and can store any type of text or binary data. It is also referred to as object storage.



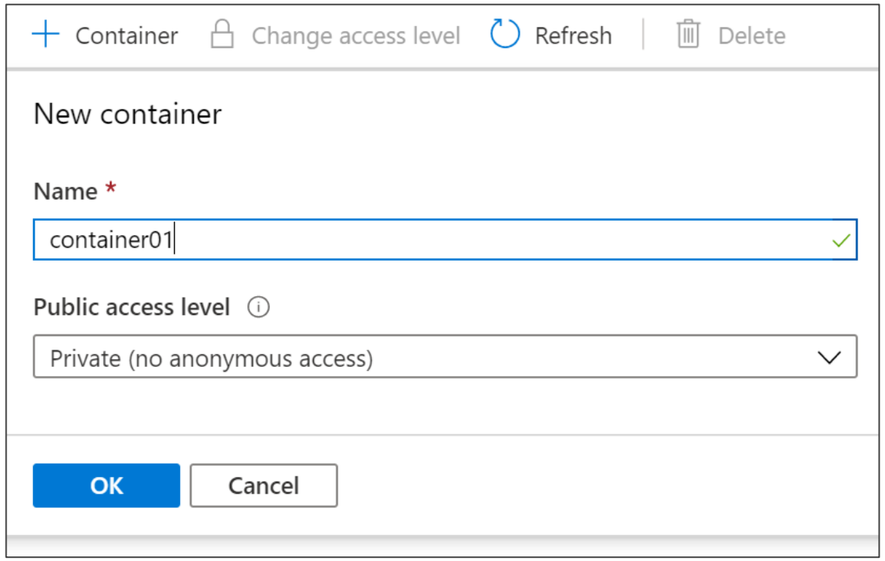
Common Uses

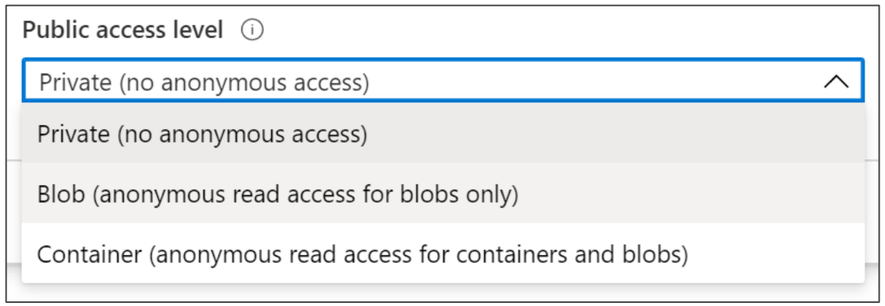
- Serving images or documents directly to a browser
- Storing files for distributed access
- Streaming video and audio
- Storing data for backup and restore, disaster recovery, and archiving
- Storing data for analysis by an on-premises or Azure-hosted service

Source: https://docs.microsoft.com/en-us/azure/?product=popular

©Simplilearn. All rights reserved.

Create Blob Storage





Guidelines to create Blob containers:

- All blobs must be in a container
- Accounts have unlimited containers
- Containers can have unlimited blobs
- Private blobs: No anonymous access
- Blob access: Anonymous public read access for blobs only
- Container access: Anonymous public read and list access to the entire container, including the blobs

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Manage Soft Delete

Azure Storage offers soft delete for the blob objects so that the user can recover data.

The soft-delete option for blob objects has the following features:

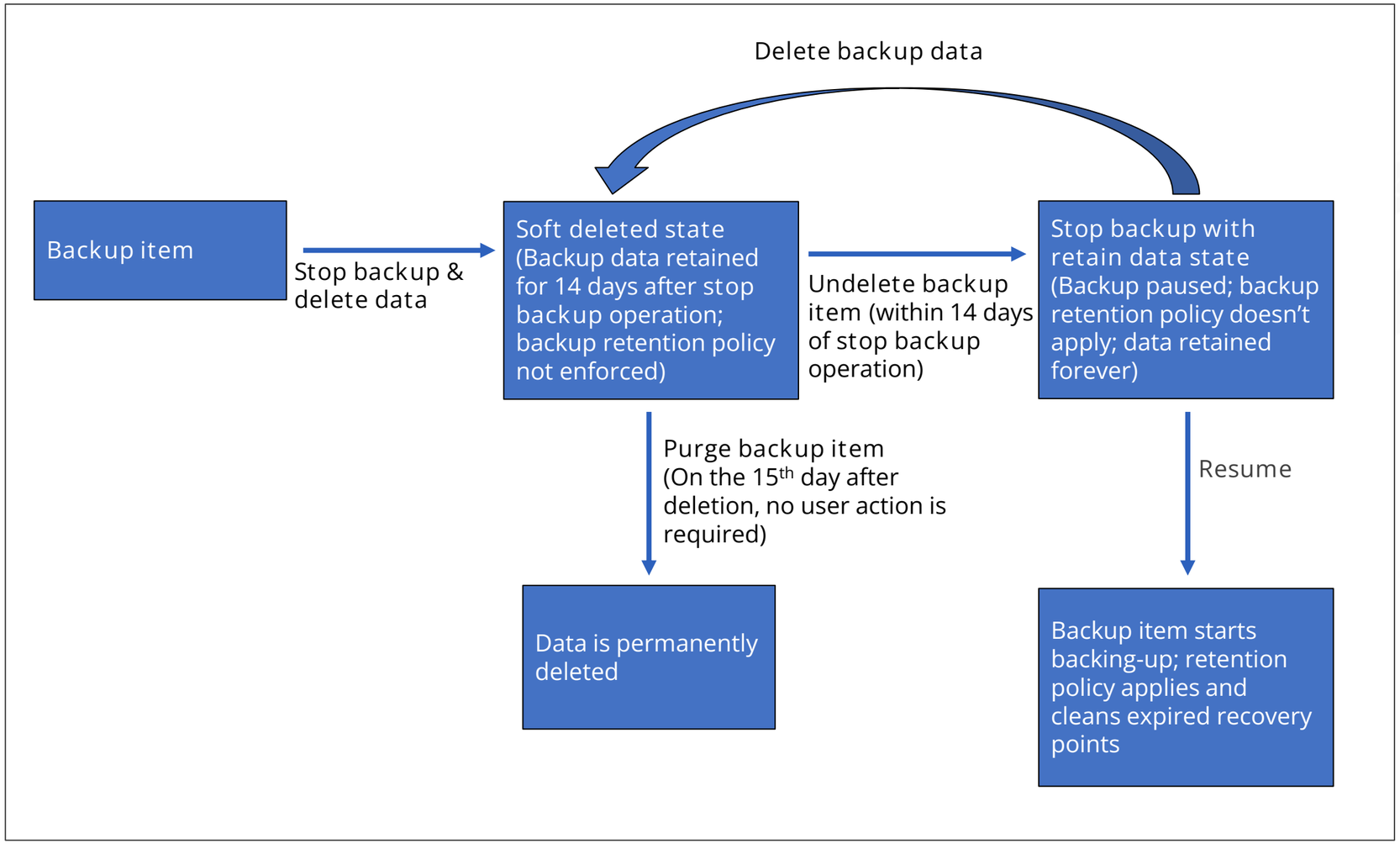
- Backup data is retained for 14 additional days.
- It recovers soft-deleted backup items using an Undelete operation.
- It is natively built-in for all the recovery services vaults.

Source: https://docs.microsoft.com/en-us/azure

©Simplilearn. All rights reserved.

Manage Soft Delete

Below image depicts the flow of managing soft delete:



Source: https://docs.microsoft.com/en-us/azure

©Simplilearn. All rights reserved.

Creating Blob Storage

Duration: 5 Min.

Problem Statement:

Create the blob container inside the storage account to hold the unstructured data.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

Steps to create blob storage:

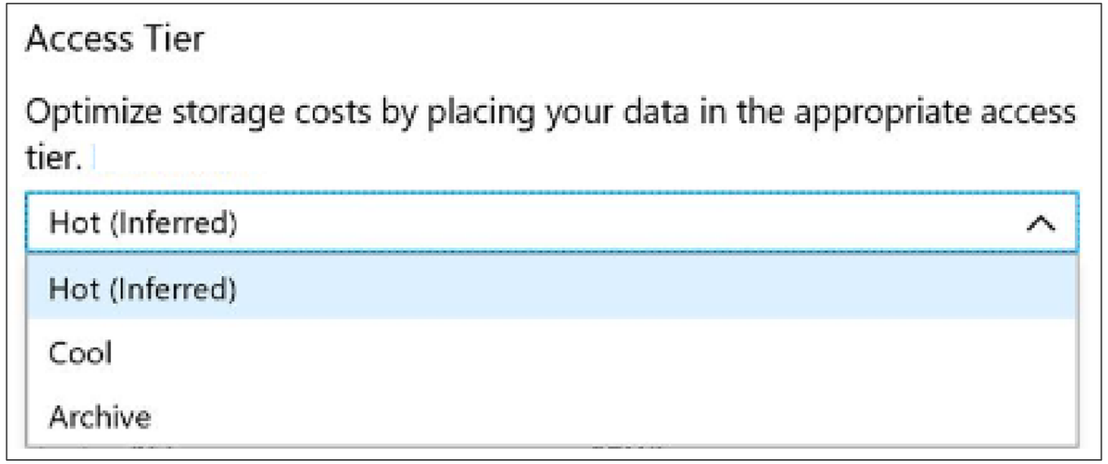
1. Go to the Azure portal
2. Create a storage account
3. Create a blob container

©Simplilearn. All rights reserved.

Create Blob Access Tiers

Access tiers by Azure Storage to access blob data:

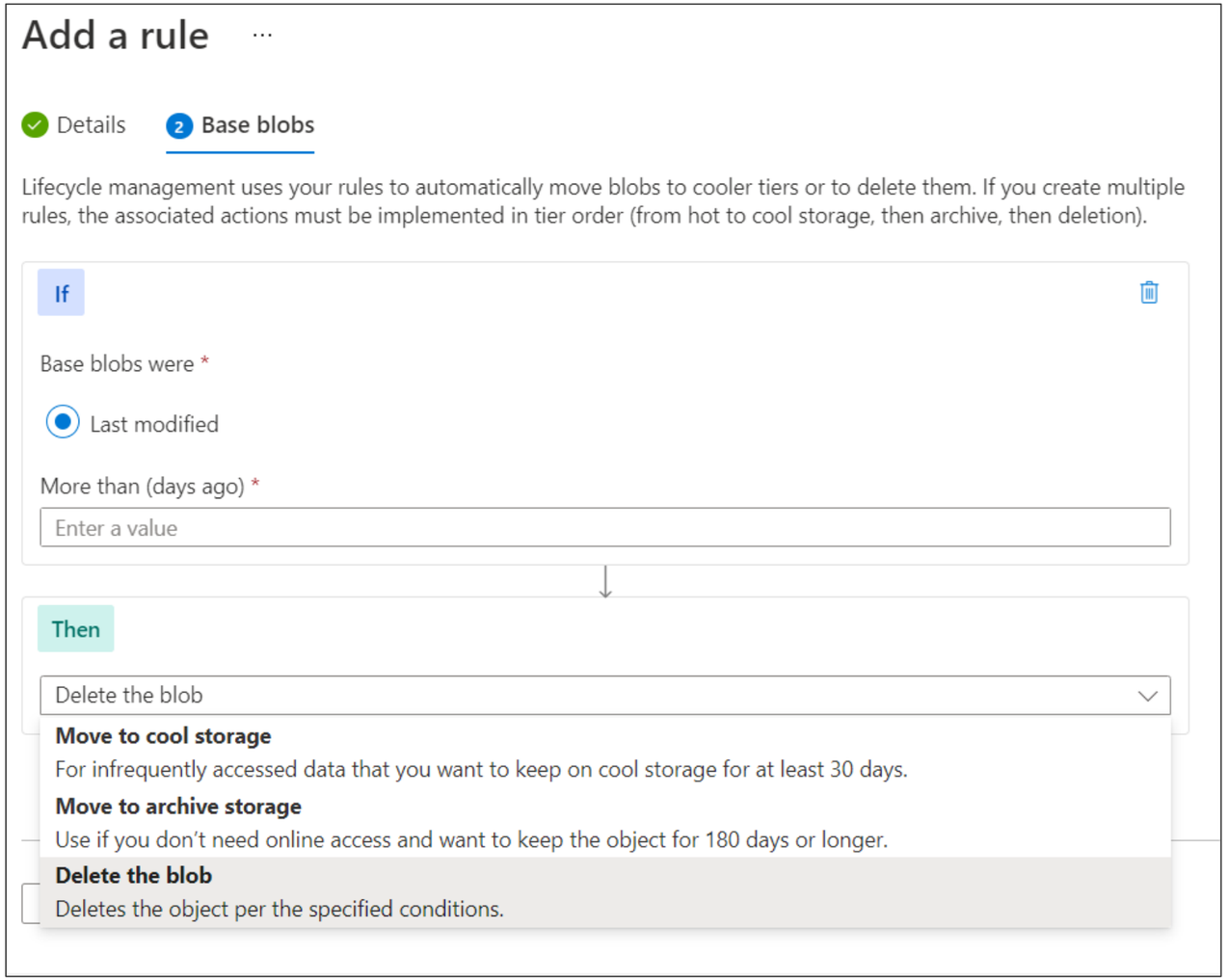
- Hot tier: Optimized for frequent access of objects in the storage account
- Cool tier: Optimized for storing large amounts of data that is infrequently accessed and stored for at least 30 days
- Archive: Optimized for data that can tolerate several hours of retrieval latency and will remain in the Archive tier for at least 180 days



Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Add Blob Lifecycle Management Rules



The lifecycle management policy allows to:

- Transition blobs to a cooler storage tier to optimize performance and cost
- Delete blobs at the end of their lifecycle
- Apply rules to filtered paths in the storage account

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Creating Blob Lifecycle Management Rules

Duration: 15 Min.

Problem Statement:

Create the Blob lifecycle management rules to store the unstructured data in the most cost-effective manner.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

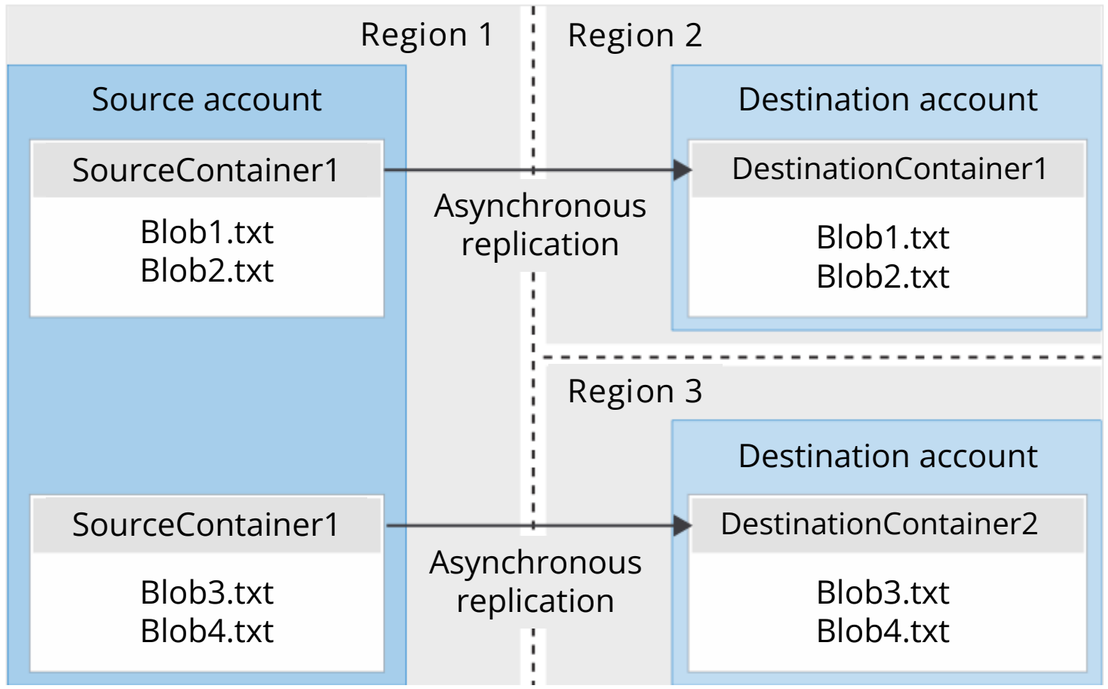
Assisted Practice: Guidelines

Steps to create Blob lifecycle management rules:

1. Go to the Azure portal
2. Locate the storage account blob
3. Select lifecycle management option

©Simplilearn. All rights reserved.

Determine Blob Object Replication



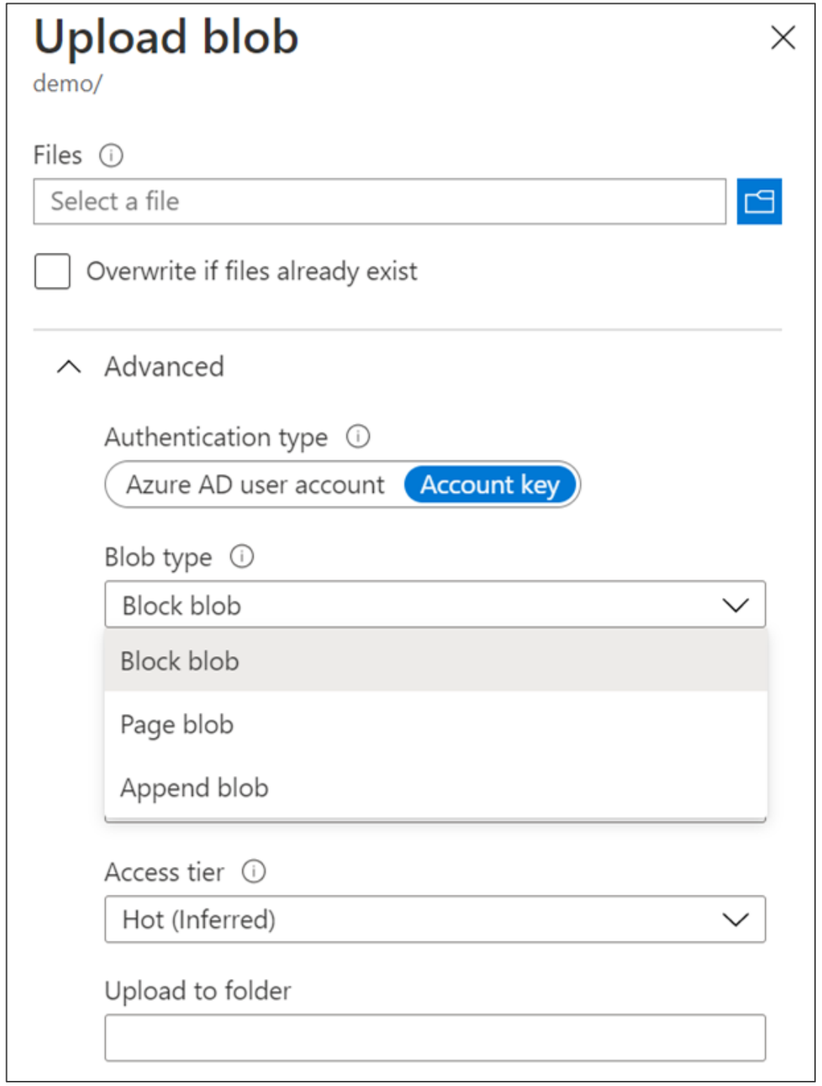
Features of Blob Object Replication:

- Is asynchronous to any other region
- Minimizes latency for reading requests
- Increases efficiency for compute workloads
- Optimizes data distribution
- Optimizes costs

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Upload Blobs



Source: https://portal.azure.com/

Authentication type for blobs:

- Azure AD user account
- Account key

Types of blobs include:

- Block blobs (default): Useful for storing text or binary files
- Page blobs: More efficient for frequent read or write operations
- Append blobs: Useful for logging scenarios

©Simplilearn. All rights reserved.

Review Storage Security Strategies

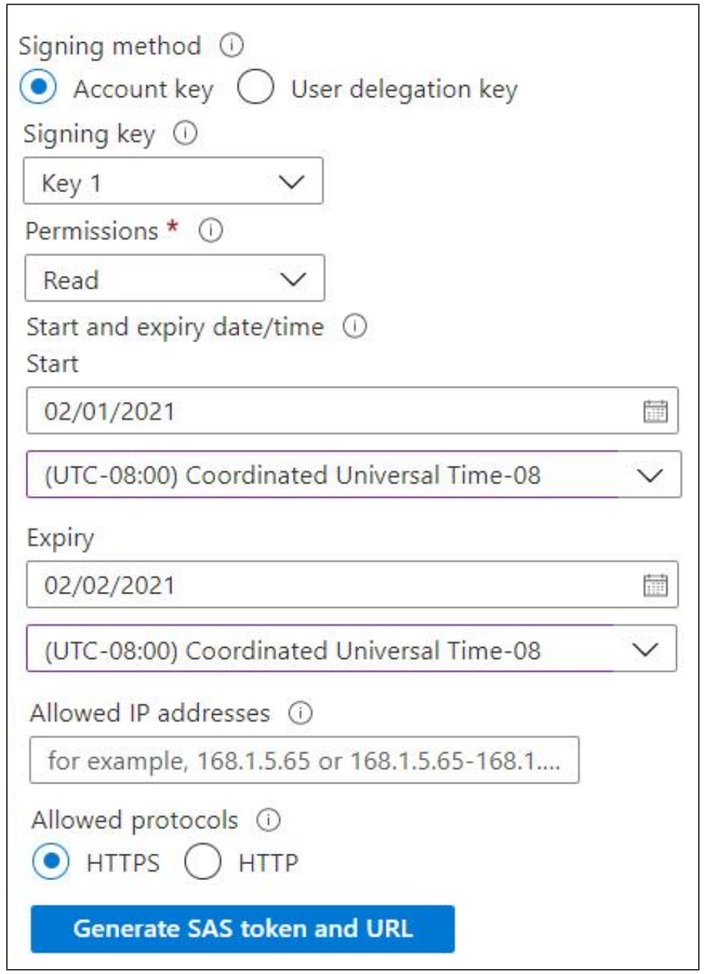
Security capabilities to build secure applications include:

- Storage service encryption
- Authentication with Azure AD and RBAC
- Client-side encryption, HTTPS, and SMB 3.0 for data in transit
- Azure disk encryption
- Shared Access Signatures: Delegated access
- Shared Key: Encrypted signature string
- Anonymous access to containers and blobs

©Simplilearn. All rights reserved.

Create Shared Access Signature

A shared access signature (SAS) is a URI that grants restricted access rights to Azure Storage resources:



- It provides delegated access to resources.
- It grants access to clients without sharing the storage account keys.
- The account SAS delegates access to resources in one or more of the storage services.
- The service SAS delegates access to a resource in just one of the storage services.

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Identify URI and SAS Parameters

- An SAS is a signed URI that points to one or more storage resources.
- It consists of a storage resource URI and the SAS token.

https://myaccount.blob.core.windows.net/?sp=r&st=2020-05-11T18:31:43Z&se=2020-05-12T02:31:43Z&spr=https&sv=2019-10-10&sr=b&sig=jOqABJZHfUVeBQ3yVn7kWiCKlO0sxCiK1rzEchfAz8U%3D

It includes parameters for the resource URI, storage services version, services, resource types, start time, expiry time, resource, permissions, IP range, protocol, and signature.

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Azure Subscriptions and Service Limits

Microsoft Azure limits are also called quotas.

Managing limits

- Some limits apply to the regional level.
- The user can raise soft limits by raising an online customer support request at no charge.
- These limits keeps on changing.
- To check the latest limits, navigate to:

  https://docs.microsoft.com/en-us/azure/azure-resource-manager/management/azure-subscription-service-limits

©Simplilearn. All rights reserved.

Creating Shared Access Signature

Duration: 10 Min.

Problem Statement:

Create a shared access signature for the storage account to provide users with secure access.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

Steps to create shared access signature:

1. Go to the Azure portal
2. Locate the storage account blade
3. Create SAS

©Simplilearn. All rights reserved.

Managing Access Levels for Azure Blob Storage

Duration: 10 Min.

Problem Statement:

You've been assigned a task to manage access levels for a blob storage container.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

Steps to be followed:

1. Select the access level for the blob storage container

©Simplilearn. All rights reserved.

Managing Firewall and Network Access

Duration: 10 Min.

Problem Statement:

You've been assigned a task to manage firewall and network access for storage accounts.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

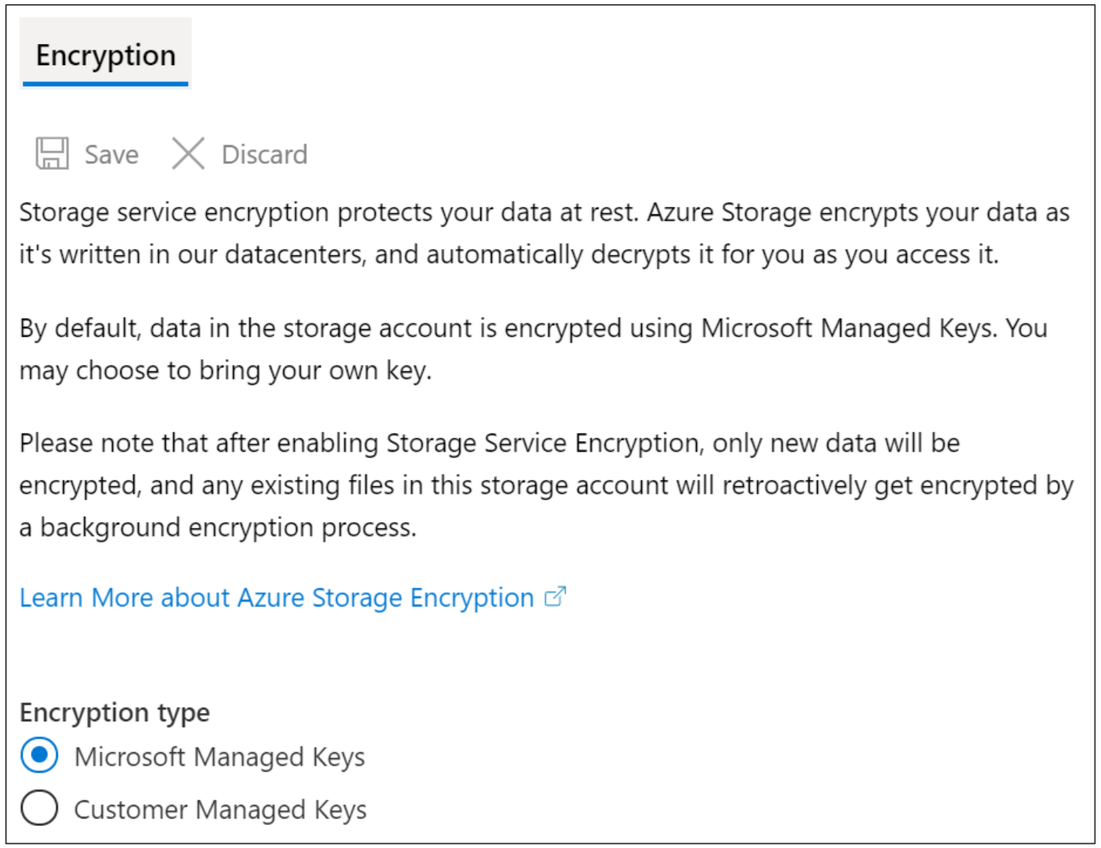
Steps to be followed:

1. Manage firewall and network access

©Simplilearn. All rights reserved.

Features of Storage Service Encryption

Azure Storage Service Encryption (SSE) for data at rest protects the data by ensuring that the organizational security and compliance commitments are met. Its features include:

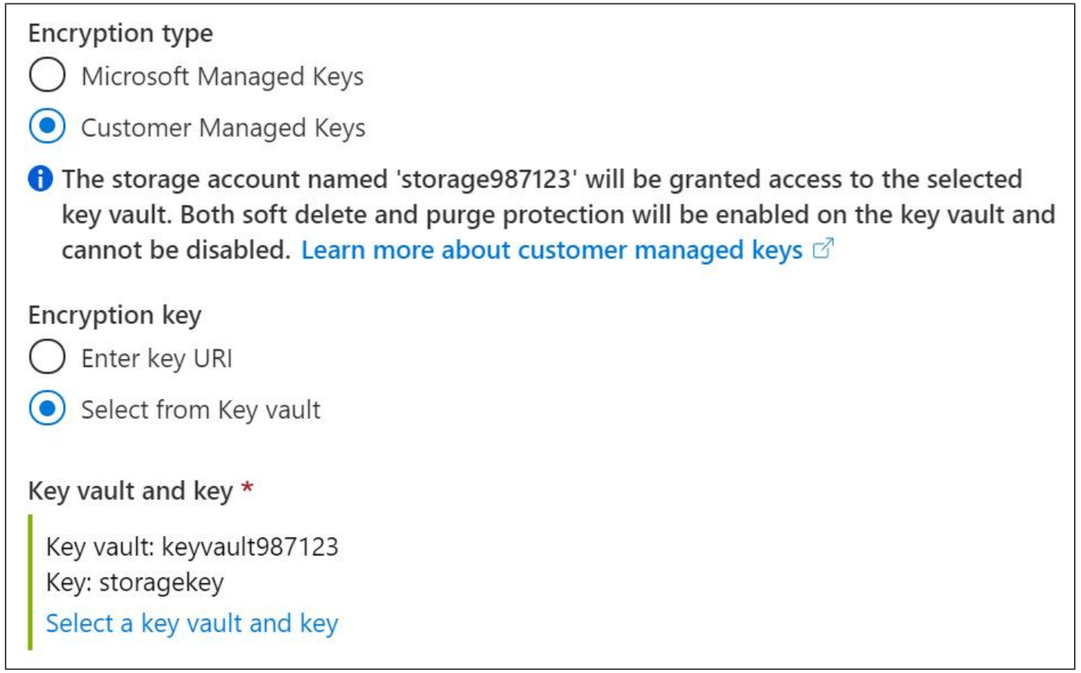


- Protects the data for security and compliance
- Encrypts and decrypts the data automatically
- Is encrypted through 256-bit AES encryption
- Is enabled for all new and existing storage accounts and cannot be disabled
- Is transparent to users

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Features of Customer Managed Keys



Features of the Customer Managed Keys:

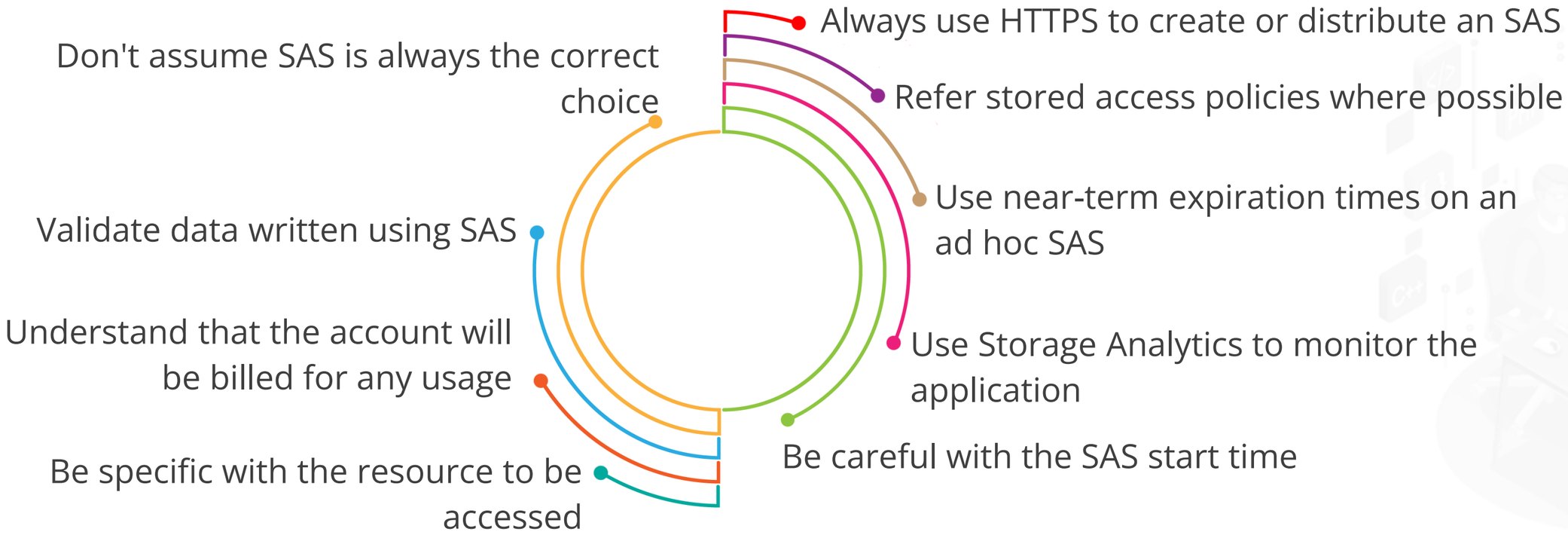
- Use the Azure Key Vault to manage the encryption keys
- Create encryption keys and store them in a key vault
- Use Azure Key Vault's APIs to generate encryption keys
- Use custom keys for more flexibility and control

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Storage Security Best Practices

The following recommendations for using shared access signatures can help mitigate risks:



©Simplilearn. All rights reserved.

Configure Azure Files

©Simplilearn. All rights reserved.

Compare Azure Files to Azure Blobs

The following table depicts the differences between Azure files and Azure blobs:

Feature	Description	When to use
Azure Files	SMB interface, client libraries, and a REST interface that allow access from anywhere to stored files	• Lift and shift an application to the cloud • Store shared data across multiple virtual machines • Store development and debugging tools that need to be accessed from many virtual machines
Azure blobs	Client libraries and a REST interface that allow unstructured data (flat namespace) to be stored and accessed at a massive scale in block blobs	• Support streaming and random-access scenarios • Access application data from anywhere

©Simplilearn. All rights reserved.

Manage File Shares

Windows Linux macOS

Drive letter

Z

To connect to this Azure file share from Windows, run these PowerShell commands from a normal (not elevated) PowerShell terminal:

```
$connectTestResult = Test-NetConnection -
ComputerName storage987123.file.core.windows.net -
Port 445
if ($connectTestResult.TcpTestSucceeded) {
    # Save the password so the drive will persist on reboot
    cmd.exe /C "cmdkey
/add:`"storage987123.file.core.windows.net`"
```

This script will check to see if this storage account is accessible via TCP port 445, which is the port SMB uses. If port 445 is available, your Azure file share will be persistently mounted. Your organization or internet service provider (ISP) may block port 445, however you may use Azure Point-to-Site (P2S) VPN, Azure Site-to-Site (S2S) VPN, or ExpressRoute to tunnel SMB traffic to your Azure file share over a different port.

Parameters for file share name and the quota:

- File share quotas
- Windows: Ensure port 445 is open
- Linux: Mount the drive
- MacOS: Mount the drive
- Secure transfer required: SMB 3.0 encryption

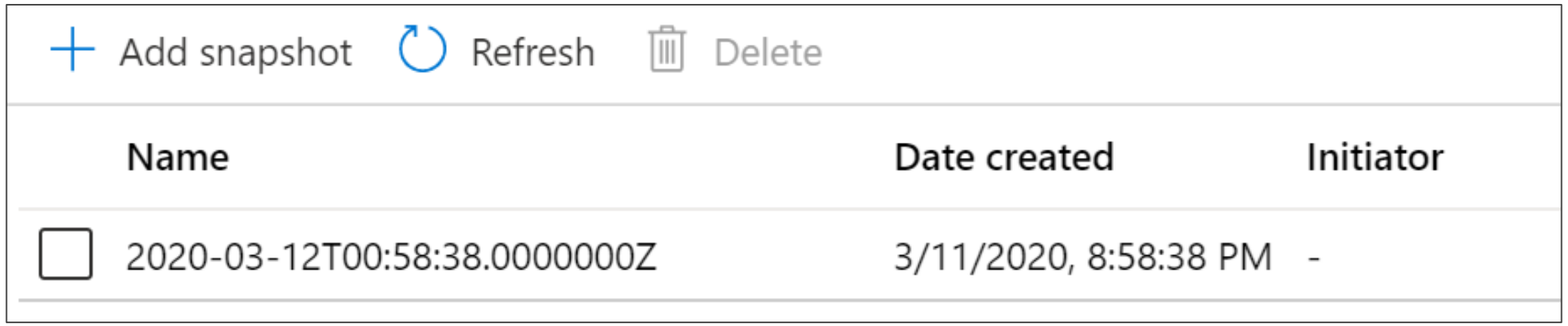
Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Create File Share Snapshots

Guidelines to create file share snapshots:

- Create an incremental snapshot that captures the shared state at a point in time and a read-only copy of the data
- Delete the snapshot at the file share level and restore it at the file level
- Protect against application error and data corruption
- Protect against accidental deletions or unintended changes
- Retain only the most recent share snapshot in order to restore the share



Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Create Azure File Share

Duration: 10 Min.

Problem Statement:

As an Azure Administrator, recommend a distributed file server service to deploy to Azure.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

Steps to create an Azure file share:

1. Go to the Azure portal
2. Navigate to the storage account
3. Select the file share option to create the file share

©Simplilearn. All rights reserved.

Create an Azure File Share Snapshot

Duration: 10 Min.

Problem Statement:

As an Azure Administrator, recommend a backup method for a distributed file server service that has been deployed to Azure.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

Steps to create an Azure file share snapshot:

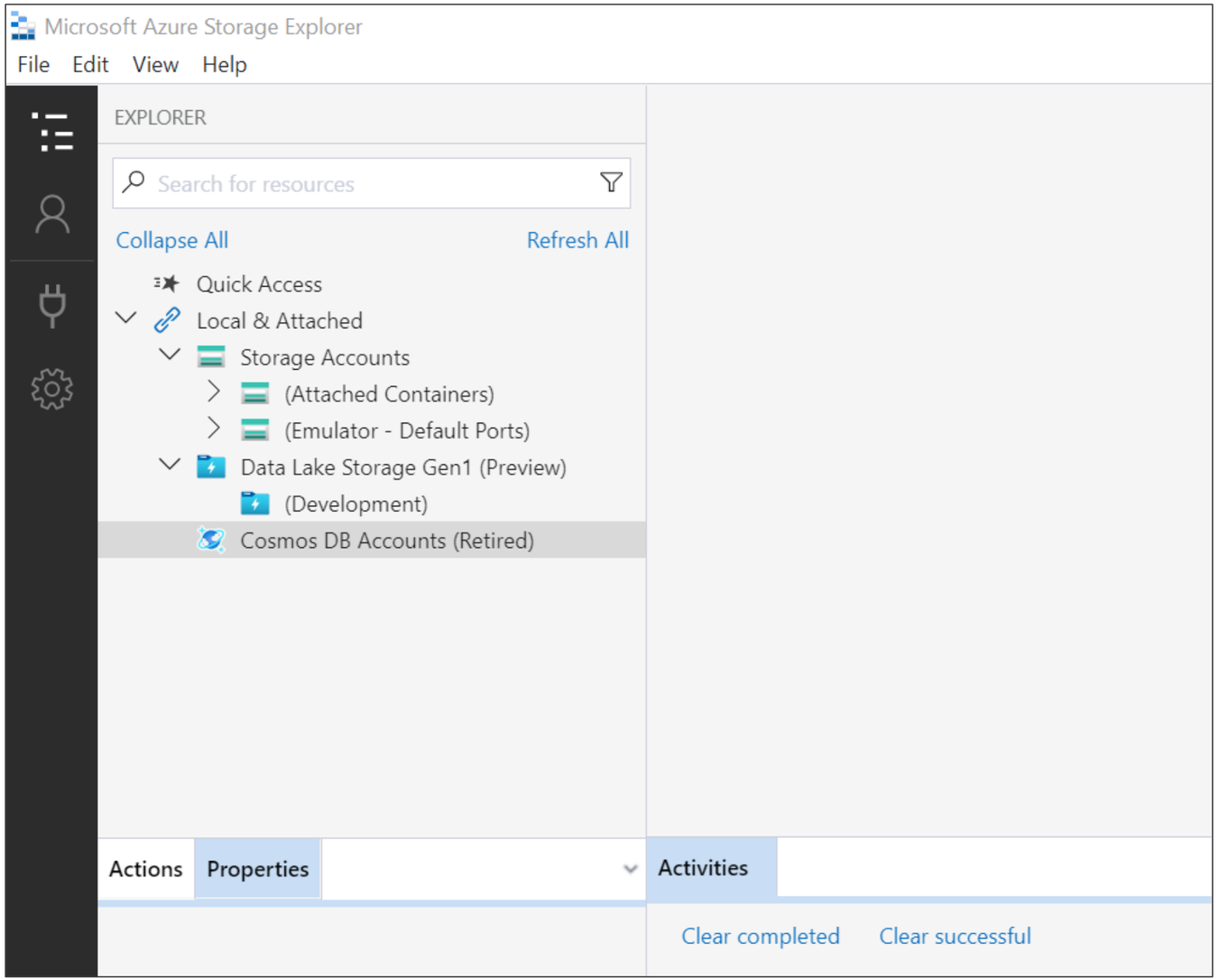
1. Go to the Azure portal
2. Navigate to the storage account
3. Locate the file share
4. Select the create snapshot option to create the snapshot

©Simplilearn. All rights reserved.

Configure Storage with Tools

©Simplilearn. All rights reserved.

Storage Explorer



Uses of Azure Storage Explorer are:

- Access multiple accounts and subscriptions
- Create, delete, view, and edit storage resources
- View and edit Blob, Queue, Table, File, Cosmos DB storage, and Data Lake Storage
- Obtain shared access signature (SAS) keys
- Available for Windows, Mac, and Linux

Source: https://portal.azure.com/

©Simplilearn. All rights reserved.

Use Azure Storage Explorer

Duration: 10 Min.

Problem Statement:

Install and explore Azure storage explorer to manage the storage account directly from local system instead of going to Azure portal.

ASSISTED PRACTICE

©Simplilearn. All rights reserved.

Assisted Practice: Guidelines

Steps to configure Azure storage explorer:

1. Install Azure storage explorer
2. Launch the explorer
3. Connect to the Azure account

©Simplilearn. All rights reserved.

Use AzCopy

An alternative method for transferring data is AzCopy.
Syntax: Azcopy copy [source] [destination] [flags]

- Command line utility
- Available on Windows, Linux, and macOS
- Designed for copying data to and from Azure blob, file, and table storage
- Authentication options include Active Directory (Entra ID) or SAS token

©Simplilearn. All rights reserved.

Key Takeaways

- The storage account provides a unique namespace for Azure Storage data objects.
- Data objects can be blobs, files, queues, tables, and disks.
- Azure Blob storage is optimized for storing massive amounts of unstructured data.
- Secure Azure storage comes with shared access signatures, storage service encryption, and customer-managed keys.

©Simplilearn. All rights reserved.

simplilearn

Key Takeaways

- Azure files offer fully managed file shares in the cloud that are accessible via SMB or NFS protocols
- Azure File Sync is a service that caches several Azure file shares on an on-premises Windows server or cloud VM.
- AzCopy is a command-line utility that can be used to copy blobs or files to or from a storage account.
- Azure Import or Export service is used to securely import large amounts of data to Azure blob storage and Azure files.

©Simplilearn. All rights reserved.

Implementing Stored Access Policy

Duration: 20 Min.

Project agenda: To implement the stored access policy

Description: Create a traffic manager to route the traffic for a web application based on performance. As a part of this, create two web applications, each in two different regions, and place them behind the traffic manager to route the traffic to appropriate applications based on performance.

Perform the following:

Create two web applications in two different regions, and use a traffic manager to route the traffic based on performance.

LESSON-END PROJECT

©Simplilearn. All rights reserved.